

Precipitation of zinc ferrite nanoparticles in the $\text{Fe}_2\text{O}_3\text{--ZnO--SiO}_2$ glass system

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Abstract

Glass materials in the $\text{ZnO--Fe}_2\text{O}_3\text{--SiO}_2$ system, containing zinc ferrite nanoparticles, were prepared by the sol–gel method and characterized using X-ray diffraction (XRD), scanning electron microscopy (SEM), Mössbauer spectroscopy, AC- and DC-magnetization techniques. The gel samples, dried at 130 °C, were further heat treated in air at 500 and 800 °C. At 500 °C zinc ferrite and hematite nanoparticles, with an average size of approximately 24 nm, were precipitated in the brown and opaque $10\text{ZnO--}10\text{Fe}_2\text{O}_3\text{--}80\text{SiO}_2$ and in the ruby colored transparent $5\text{ZnO--}5\text{Fe}_2\text{O}_3\text{--}90\text{SiO}_2$ and $2.5\text{ZnO--}2.5\text{Fe}_2\text{O}_3\text{--}95\text{SiO}_2$ glass matrices. In the $5\text{ZnO--}5\text{Fe}_2\text{O}_3\text{--}90\text{SiO}_2$ sample the nanoparticles exhibited ferro or ferrimagnetic interactions combined with superparamagnetism with a blocking temperature of approximately 14 K. Heating at 800 °C seems to cause partial dissolution of the zinc ferrite and hematite particles in all the investigated compositions. Accordingly at 800 °C the $5\text{ZnO--}5\text{Fe}_2\text{O}_3\text{--}90\text{SiO}_2$ glass shows a paramagnetic behavior down to 2 K.

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1. Introduction

Glass–ceramics with a bioactive glass matrix and containing ferrimagnetic, ferromagnetic or superparamagnetic particles are potentially useful in bone tumor treatment [1]. The preparation of these bioglass-ceramics by the sol–gel method, a flexible low-temperature process, allows an easy control of the structure and of the particle size necessary to produce materials with a well-defined magnetic response for specific biomedical applications. The use of sol–gel silica matrices as dispersion media for magnetic nanoparticles has a further advantage. A surface enriched in silica leads to the presence of surface silanol groups that can easily react with alcohols and silane coupling agents to produce

dispersions that are not only stable in non-aqueous solvents but also provide the ideal anchorage for covalent bonding of specific ligands [2].

The magnetic properties of the zinc ferrite ZnFe_2O_4 are highly affected by their particle size [3]. Bulk ZnFe_2O_4 is an antiferromagnetic material that crystallizes in the normal spinel structure which contains two different cation sites: eight tetrahedral A sites and 16 octahedral B sites per f.u. Magnetic coupling occurs via super exchange between the Fe^{3+} cations. The A–A and B–B couplings are weak compared to the A–B interaction. In a normal spinel all Zn^{2+} cations are on A sites and all Fe^{3+} cations on B sites. In a fully inverse spinel half of the trivalent cations fully occupy the A sites while the B sites are shared by the divalent cations and the remaining half of the trivalent cations. Mixtures between the two configurations occur and are characterized by the degree of inversion which depends strongly on the preparation procedures [4,5]. The zinc

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ferrite nanoparticles may have non-zero inversion degrees which leads to different magnetic properties. Namely, they may show ferromagnetism or ferrimagnetism combined with superparamagnetism [3]. The growth of nanoparticles in a silica matrix prevents their coarsening, aggregation and interaction. Several studies have shown that the inter-particle interaction can have a strong influence on the magnetic properties of the samples [6].

The aim of this work is the preparation and characterization of silica sol–gel derived glass–ceramics containing nanoparticles of zinc ferrite. The samples were studied using X-ray diffraction (XRD), scanning electron microscopy (SEM), Mössbauer spectroscopy, AC and DC-magnetization techniques.

2. Experimental

2.1. Sample preparation

The glass samples with the compositions indicated in Table 1 were prepared using tetraethylorthosilicate (TEOS), zinc and iron nitrates. A mixture of TEOS, ethanol, and zinc and iron nitrates dissolved in HCl acidified water (acidified water/TEOS molar ratio of 10) was stirred during one hour at room temperature. Solutions were allowed to gel and dry at 50 °C in Petri dishes. The obtained gels were heat-treated at 130 °C. These 130 °C treated samples were further heated at 500 °C and 800 °C, under ambient air conditions.

2.2. Sample characterization

The samples were characterized by powder X-ray diffraction, XRD, and scanning electron microscopy, SEM. The particle average size was calculated based on the XRD powder patterns and using the Scherrer equation $D = 0.9\lambda/\beta\cos\theta$ where λ stands for the radiation wavelength, β the size contribution to the integral breadth of a diffraction peak and D the apparent size (in Å) along the corresponding reciprocal lattice vector. The X-ray diffractometer and scanning electron microscope were a Rigaku XDMAX and a Hitachi S4100-1, respectively.

Mössbauer spectra were measured in transmission mode using a conventional constant-acceleration spectrometer and a 25 mCi ^{57}Co source in a Rh matrix. The velocity scale was calibrated using an $\alpha\text{-Fe}$ foil at room temperature. Absorbers were obtained by pressing the sample powder (5 mg of natural Fe/cm^2) into a perspex holder. Spectra collected below room temperature were obtained with the

sample in exchange He gas flow using a cryostat with temperature stability of approximately 0.5 K. The spectra were analyzed using a non-linear least-squares computer method [4]. Center shifts, CS, are given relative to metallic $\alpha\text{-Fe}$ at room temperature. Labeling the six lines observed in a magnetic sextet from 1 to 6, from negative to positive velocities, the calculated widths, Γ , and relative areas of both peaks in peak pairs 1–6, 2–5 and 3–4 in each magnetic sextet, were constrained to remain equal during the fitting procedure. The calculated Γ and the relative areas of both peaks in each quadrupole doublet were also constrained to remain equal during refinement. Distributions of quadrupole splittings or magnetic splittings were fitted according to the histogram method [7].

The magnetic measurements, AC-susceptibility and DC-extraction were performed on powder samples (10–15 mg) using a multipurpose characterization system, *MagLab 2000* (Oxford Instruments) in the temperature range 1.7–300 K and with magnetic fields up to 12 T. DC-magnetization were recorded under zero-field cooled (ZFC) and field-cooled (FC) sequences under 0.05 and 0.1 T. Magnetic hysteresis loops were recorded at several temperatures below and above the blocking temperature. Frequency-dependent ($f = 10\text{ Hz to } 10\text{ kHz}$) AC-susceptibility measurements, $\chi_{\text{AC}}(f, T)$, were obtained using an AC-amplitude of 0.1 mT. For a single frequency (995 Hz), external DC-fields of 5, 10 and 20 mT, were also applied during the thermal scans.

3. Results and discussion

3.1. Samples heat treated at 500 °C

The 20ZnFe sample (Table 1) treated at 500 °C was found to be brown and opaque. The corresponding XRD pattern shows the presence of two crystalline phases, hematite and zinc ferrite (Fig. 1). The average size of the hematite and zinc ferrite particles, determined by the Scherrer method, is approximately 24 nm and 80 nm, respectively. In the SEM micrographs of this sample the presence of numerous particles may also be observed (Fig. 2). However, in the micrograph of the fracture surface, representative of the bulk of the sample, the particles are not well defined. The particles observed by SEM in the free surface of the sample (Fig. 2a) have a size larger than that calculated by the Scherrer method. This may be explained by the thermodynamic conditions in the surface of the samples which propitiate the growth of crystals. The vast majority of crystalline zinc ferrite and hematite contributing to the XRD patterns are in the bulk of the samples and therefore with average particle sizes much smaller than those observed at their surface. The same phenomenon is observed in the 10ZnFe-500 °C and 5ZnFe-500 °C ruby samples (Fig. 3). While in the free surface of the 10ZnFe-500 °C sample cubic particles with an average size of 140 nm as well as some spherical particles are observed

Table 1
Samples composition (mole%)

Sample	SiO_2	$\text{ZnO} + \text{Fe}_2\text{O}_3$
5ZnFe	95	5
10ZnFe	90	10
20ZnFe	80	20

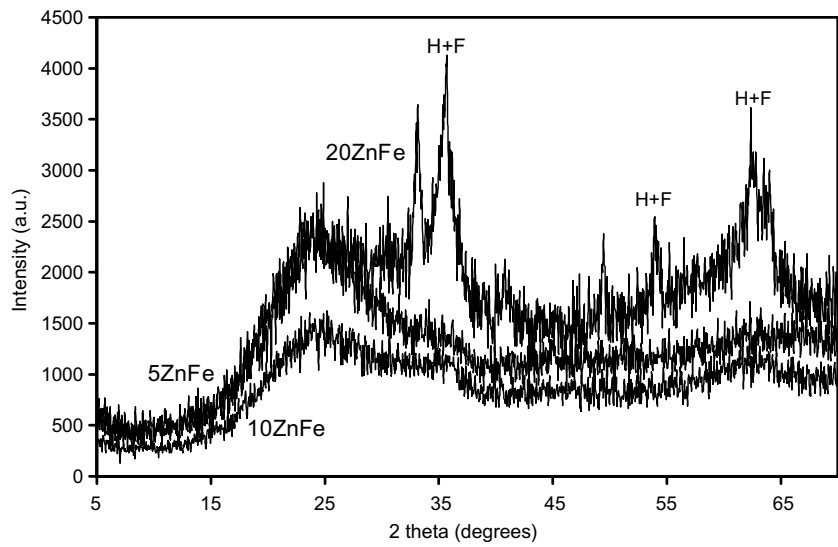


Fig. 1. XRD patterns of the 20ZnFe, 10ZnFe and 5ZnFe samples treated at 500 °C. H – hematite; F – zinc ferrite.

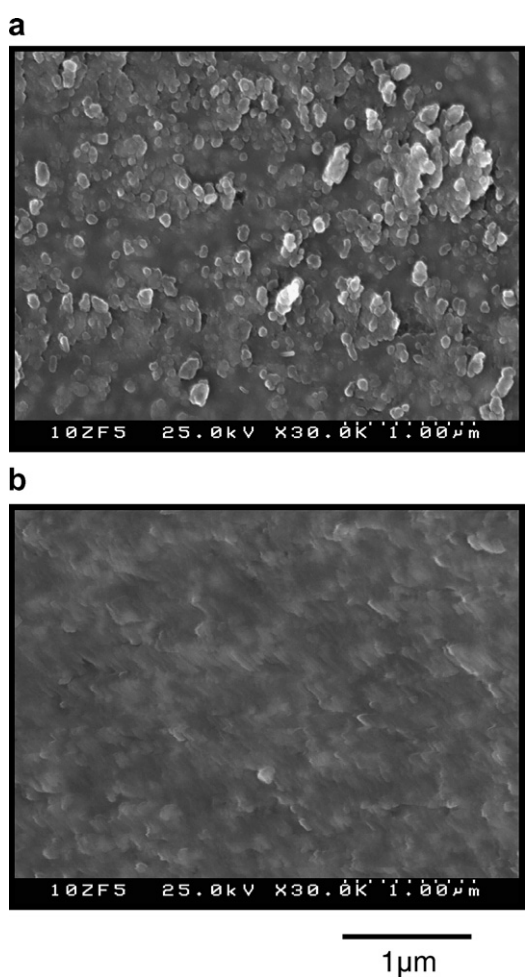


Fig. 2. SEM micrographs of 20ZnFe-500 °C. a – free surface; b – fracture surface.

(Fig. 3a), in the fracture surfaces none of these particles are visible (Fig. 3b).

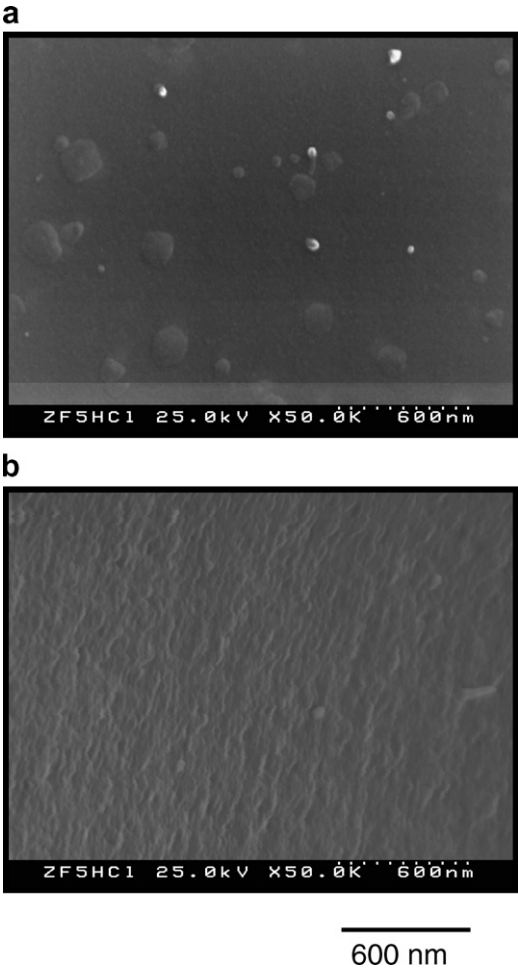


Fig. 3. SEM micrographs of 10ZnFe-500 °C sample. a – free surface; b – fracture surface.

The XRD pattern and the fracture surface micrograph of the 10ZnFe-500 °C sample (Figs. 1 and 3b) do not

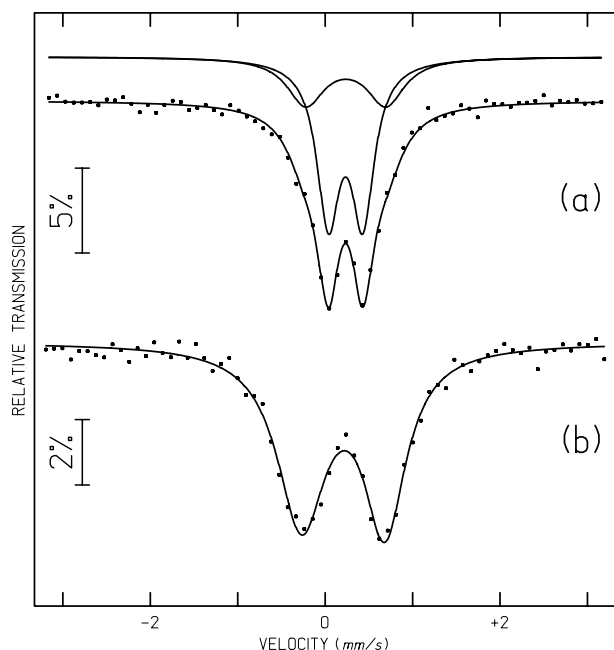


Fig. 4. Room temperature Mössbauer spectra of the glass samples heat-treated in air at (a) 500 °C and (b) 800 °C. The total function plotted on the experimental points is the sum of the calculated doublets (shown slightly shifted, for clarity) as explained in the text. Estimated parameters in Table 2.

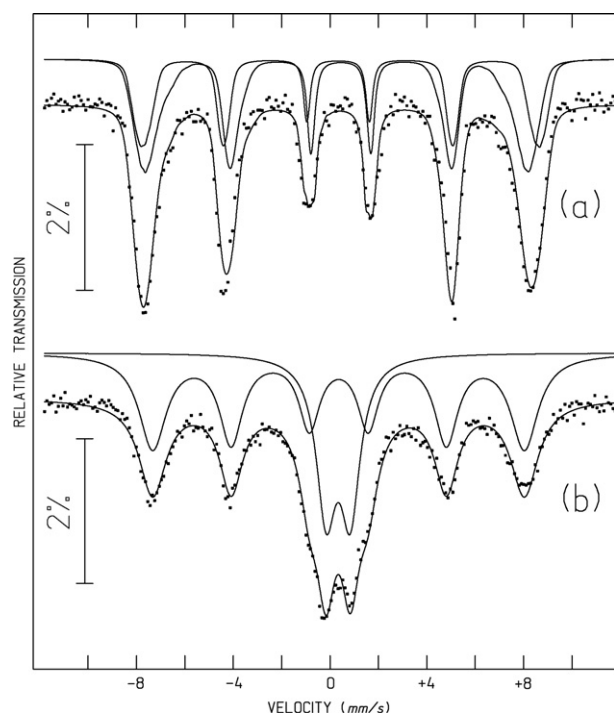


Fig. 5. Mössbauer spectra taken at 4 K of the glass samples heat-treated in air at (a) 500 °C and (b) 800 °C. Estimated parameters in Table 2.

clearly reveal the presence of hematite or zinc ferrite particles. Nevertheless, in the XRD pattern a large band centered near 2θ approximately 35.6° may be associated to the presence of zinc ferrite and hematite nano-crystalline phases.

As shown below Mössbauer spectroscopy and magnetization measurements (Figs. 4–10), confirm that zinc ferrite is present in the 10ZnFe-500 °C sample. Thus, in the room temperature Mössbauer spectrum of this sample (Fig. 4a) the presence of two shoulders, at the base of the two main absorption peaks, indicates that at least two quadrupole doublets have to be considered in the analysis. At 4 K the asymmetrical broadening of the six absorption peaks in the spectrum (Fig. 5a) has to be analyzed considering at least two magnetic sextets. If Lorentzian line shapes are assumed the refinement of the experimental data leads to much broader line-widths for peaks 1–6 than for peaks 2–5 and 3–4 of each sextet. Such a strong broadening of the external peaks of a sextet as compared to the inner ones suggests the presence of a distribution of magnetic hyperfine fields B_{hf} . This distribution may be easily explained by the different neighborhoods of the Fe atoms in the

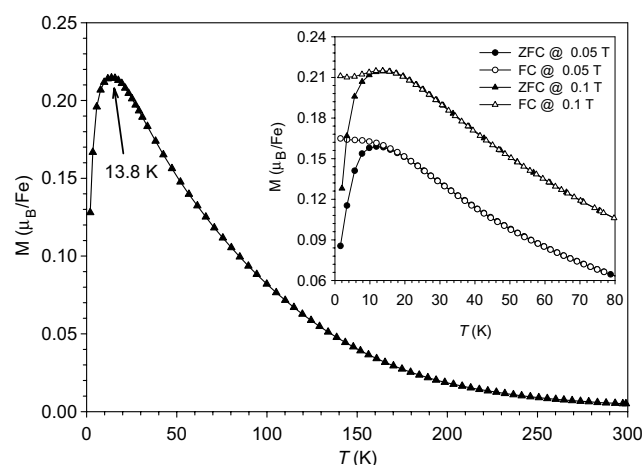


Fig. 6. Temperature dependence of the magnetization, $M(T)$ of the 10ZnFe-500 °C sample in zero field cooled (ZFC) and field cooled (FC), measured at 0.05 and 0.1 T. The inset shows in detail the shift of the peaks for both applied fields.

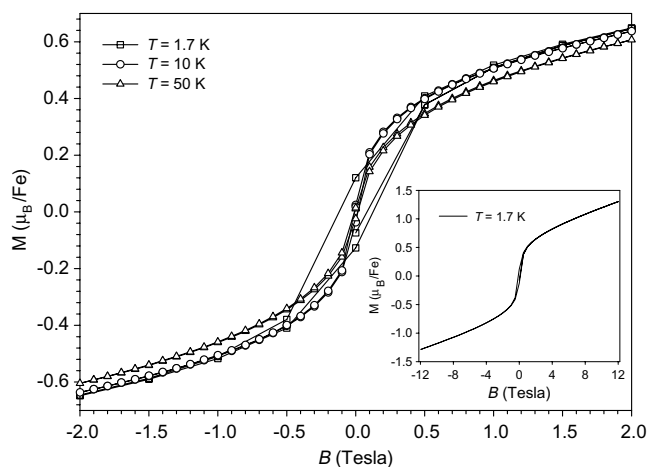


Fig. 7. Magnetic field dependence of the magnetization, $M(B)$, of the 10ZnFe-500 °C sample, at various temperatures, 1.7, 10 and 50 K. The inset shows the isothermal curve at 1.7 K up to 12 T.

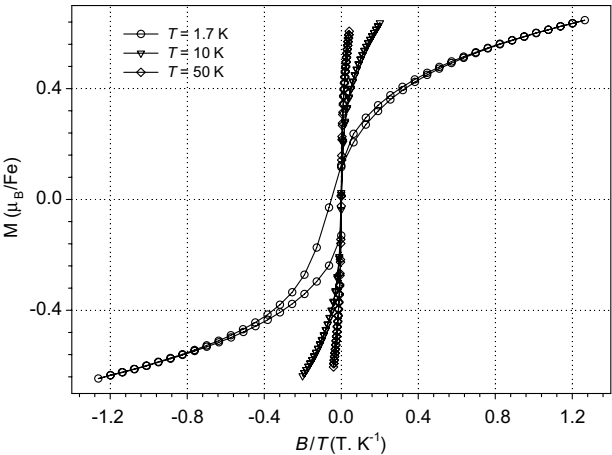


Fig. 8. Magnetization versus B/T , $M(B/T)$, of the 10ZnFe-500 °C sample taken at different temperatures, 1.7, 10 and 50 K.

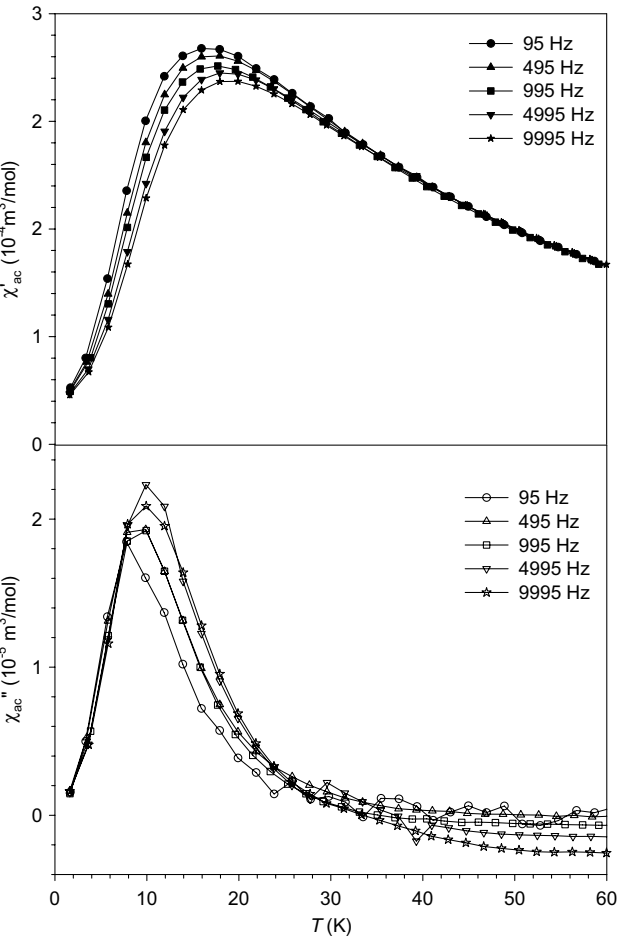


Fig. 9. Temperature dependence of the in phase $\chi'(T)$ and the out-of phase, $\chi''(T)$ components of the AC susceptibility of the 10ZnFe-500 °C sample, taken at frequencies 95, 495, 995, 4995 and 9995 Hz with an AC-amplitude of 0.1 mT.

amorphous structure of the glass sample. The 4 K spectrum of the 10ZnFe-500 °C sample was therefore fitted considering two B_{hf} distributions. One of the possible fits with the

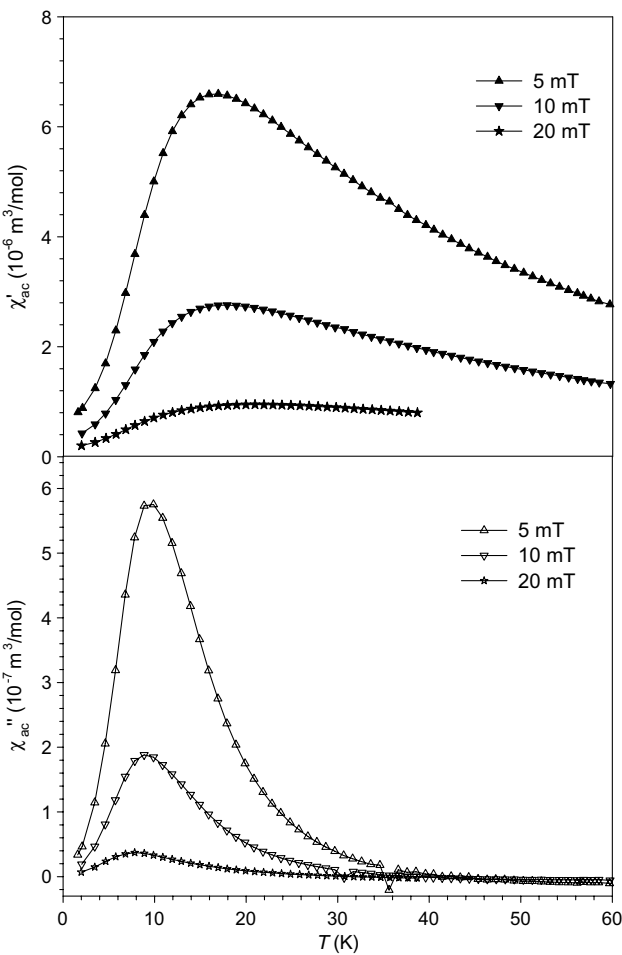


Fig. 10. Temperature dependence of the in phase $\chi'(T)$ and the out-of phase, $\chi''(T)$ components of the AC susceptibility of the 10ZnFe-500 °C sample, taken with an AC amplitude of 0.1 mT and external DC-magnetic fields of 5, 10 and 20 mT.

lowest χ^2 values (Table 2) leads to relative distribution areas similar to those estimated for the doublets at room temperature. The estimated CS and average magnetic

Table 2
Parameters estimated from the Mössbauer spectra taken at 295 and 4 K of the glass samples heat-treated in air at different temperatures T_{an}

T_{an} (°C)	T (K)	CS (mm/s)	QS (mm/s)	ε (mm/s)	$\langle B_{\text{hf}} \rangle$ (T)	B_{hf} (T)	A (%)
500	295	0.39	0.40	—	—	—	70
		0.41	0.93	—	—	—	30
500	4	0.47	—	−0.09	48.2	—	64
		0.50	—	0.05	50.7	—	36
800	295	0.32	0.95	—	—	—	100
		0.45	1.01	—	—	—	33
800	4	0.47	—	−0.01	—	47.6	67

CS centre shift relative to metallic α -Fe at 295 K. QS quadrupole splitting. $\varepsilon = (e^2 V_{\text{ZZ}} Q/4)(3 \cos^2 \theta - 1)$ is the quadrupole shift with θ being the angle between B_{hf} and the main component V_{ZZ} of the electric field gradient tensor. B_{hf} magnetic hyperfine field. $\langle B_{\text{hf}} \rangle$ average magnetic hyperfine field of a B_{hf} distribution. A relative area. Estimated errors for CS, QS, and ε are <0.02 mm/s, for $B_{\text{hf}} < 0.2$ T and for $A \leq 3\%$.

hyperfine fields $\langle B_{\text{hf}} \rangle$ are within experimental error close to those reported for ZnFe_2O_4 [8]. The Mössbauer spectra suggest therefore the presence of octahedrally and tetrahedrally coordinated Fe^{3+} . The contribution of minor hematite that might be present would overlap the sextets attributed to ZnFe_2O_4 . Although accurate amounts of Fe^{3+} in each coordination may not be obtained, due to the strong overlapping of both contributions to the spectra, the spectra of 10ZnFe -500 °C are consistent with the presence of ZnFe_2O_4 with a non-zero inversion degree, as expected for nano-sized particles of this ferrite [3,8], and do not discard the presence of hematite particles.

The DC-magnetization data of the 10ZnFe -500 °C sample were taken as ZFC/FC curves for applied magnetic fields of 0.05 and 0.1 T (Fig. 6). The ZFC magnetization curve, M_{ZFC} , shows a well defined broad peak that is centered at approximately 14 K and below which irreversible behavior sets in. This broad peak may be associated with the superparamagnetic blocking temperature, T_B , of the nanoparticles detected by XRD (Scherrer method). The observed behavior is due to the progressive magnetic blocking of these particles whose size determines a magnetic anisotropy comparable to thermal energy [9,10]. The bifurcation point between the FC and ZFC curves shifts to lower T as the magnetic field used for the measurements increases as also expected for superparamagnetic behavior [11–13]. In spin glasses, the FC magnetization curve, M_{FC} , for the temperatures below the bifurcation point is almost flat but in our data the magnetization continues to rise, which is characteristic of a superparamagnetic system [11]. At high temperatures, $T > 220$ K, the magnetization follows approximately a Curie–Weiss (CW) law. Fitting the data with the expression $\chi_{\text{DC}} = C/(T - \theta)$ an effective moment $\mu_{\text{eff}} = 5.1\mu_B/\text{Fe}$ was obtained, within the typical range for high spin, $S = 5/2$, Fe^{3+} ($\mu_{\text{eff}} = 5.9\mu_B$).

In Fig. 7 the magnetic field dependence of the magnetization at various temperatures is shown above and below T_B . The relaxation time increases with decreasing temperature. For the 10ZnFe -500 °C sample, hysteresis, due to the slow response of the system, is only present at 1.7 K well below T_B . With an applied field of 12 T magnetization reaches a value of $1.3\mu_B/\text{Fe}$ and a coercive field of approximately 0.125 T is observed. No saturation is found up to 12 T. For a measurement taken at 10 K, close to T_B , no coercive field is detected. These observations are characteristic of both superparamagnetic and spin glass systems [14,15]. The measurements of the isothermal $M(B)$ curves in both ZFC and FC conditions show that the loops are identical and symmetric about their center which is characteristic of superparamagnetic behavior [11,16]. Additionally, there is no superposition of $M(B/T)$ curves for the different temperatures (Fig. 8) as expected for a system of non-interacting superparamagnetic particles [13].

The in-phase component, $\chi'(T)$, of the temperature dependence of χ_{AC} taken at several frequencies show a frequency dependent broad peak (Fig. 9). The temperature

values where the maxima of these peaks are observed increase with increasing frequency. Near the maximum of the $\chi'(T)$ peaks the out-of-phase component, $\chi''(T)$, becomes non-zero. The peak height also decreases with the increasing frequency. This particular behavior is indicative of slow relaxation processes that are found in glassy materials, both superparamagnets and spin glasses where the freezing into a non-equilibrium state takes place at the frequency dependent temperature T_B . In order to distinguish between these two systems the equation proposed by Mydosh [14]

$$T_{\text{shift}} = T_{\text{max}}/[T_{\text{max}} \cdot \Delta \log_{10}(\omega)],$$

was applied to our results. The relative shift in T_B per decade of frequency is found to be approximately 0.12, which fits well within the range of T_{shift} values predicted for non-interacting fine particle systems, characteristic of superparamagnets and clearly higher than the typical range for spin glasses. This agrees with the above referred features of $M(T)$ and $M(B/T)$ curves which also suggest a superparamagnetic system.

The frequency dependence of T_B was also fitted to the Arrhenius law,

$$\omega^{-1} = \tau_0 \cdot \exp \cdot (E_a/(K_B \cdot T_{\text{max}})),$$

where τ_0 is the pre-exponential relaxation time and E_a/K_B is the activation energy. For superparamagnetic particles, τ_0 is usually between 10^{-9} and 10^{-13} s [14,17]. Fitting this equation to the in-phase $\chi'(T)$ data, the parameters $\tau_0 = 6.5 \times 10^{-13}$ s and $E_a/K_B = 373$ K are obtained, which corroborate the results described above. The AC-susceptibility was also recorded for a number of different applied external DC-fields in order to determine the evolution of the transition with the magnetic field (Fig. 10). The effect of small magnetic fields results in a suppression of the peak in both $\chi'(T)$ and $\chi''(T)$ components, which is again consistent with superparamagnetic behavior.

While bulk ZnFe_2O_4 orders antiferromagnetically at 10 K, nano zinc ferrite particles exhibit ferro- or ferrimagnetic interactions with higher ordering temperatures combined with superparamagnetic behavior [3]. The superparamagnetic behavior above deduced from the AC and DC-magnetization data is therefore assigned to the partially inverted nano zinc ferrite particles detected by XRD and Mössbauer spectroscopy.

3.2. Samples heat treated at 800 °C

The XRD patterns of all the samples treated at 800 °C (Fig. 11) show the presence of zinc ferrite particles with an average size, determined by the Scherrer method, of approximately 6 nm in the 5ZnFe and 10ZnFe samples, with a ruby color and 21 nm in the 20ZnFe sample which is brown and opaque. As in the case of the 500 °C samples, the particles observed in the free surface of the samples by SEM (Fig. 12–14), have larger dimensions than the average

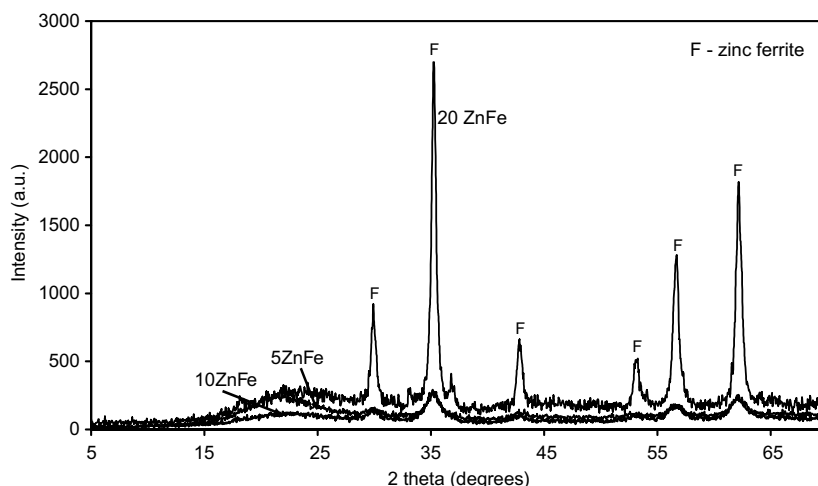


Fig. 11. XRD patterns of the 20ZnFe, 10ZnFe and 5ZnFe samples treated at 800 °C. F – zinc ferrite.

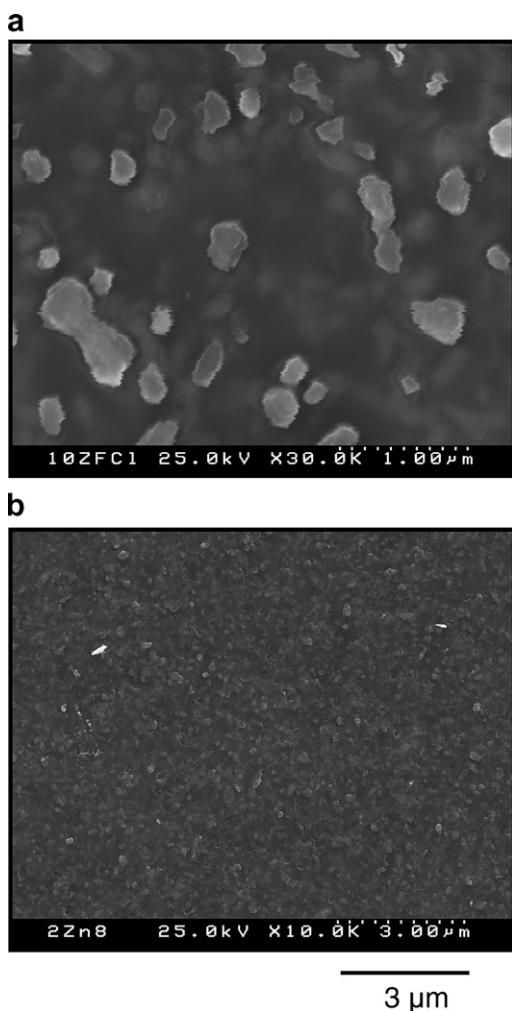


Fig. 12. SEM micrographs of 20ZnFe-800 °C sample. a – free surface; b – fracture surface.

size calculated by the Scherrer method while in the fracture surface they are difficult to observe.

The cubic particles observed in the free surface of the 10ZnFe-500 °C and 800 °C samples (Figs. 3 and 13) have similar morphology. However, the particles at 800 °C, with rounded edges and apparently thinner in the borders than in the central part, seem to be in a dissolution process. The room temperature Mössbauer spectrum of the 10ZnFe-800 °C sample (Fig. 4b) may be fitted by only one doublet. At 4 K (Fig. 5b) a doublet is still present. A sextet with very broad peaks may also be observed suggesting that the relaxation of the direction of part of the Fe magnetic moments is slowing down reaching frequencies whose reciprocal values are similar to the typical observation time of the Mössbauer effect. The broadening of the sextet peaks indicates that the relaxation frequencies are not as low as those observed for all Fe^{3+} cations in the 10ZnFe-500 °C sample at 4 K. These data show that T_B of the nanoparticles in the 800 °C sample has significantly decreased relative to that observed for the 500 °C sample, in agreement with the decreasing size detected by XRD and the dissolution effect observed by SEM. Thus, the average dimension of the particles is lower at 800 °C than at 500 °C. The magnetization data confirms also the decreasing in the average size of the particles. Contrary to the results obtained for the sample annealed at 500 °C, the magnetic measurements performed on the sample annealed at 800 °C show typical results of a simple paramagnetic system. The temperature dependence of the magnetization $M(T)$ at $B = 0.1$ T shown in Fig. 15 suggests that the compound is paramagnetic in the whole 300–1.7 K range since no anomalies are observed. The inverse of the magnetic susceptibility curve, $\chi^{-1}(T)$, is almost linear in the 1.7–250 K range (inset of Fig. 6) and the high temperature data can be fitted rather well to a modified Curie–Weiss law (MCW),

$$\chi = \chi_0 + C/(T - \theta)$$

with $\theta = 4.9$ K and $\mu_{\text{eff}} = 1.3\mu_B/\text{Fe}$. The positive Curie temperature suggests the presence of ferro- or ferrimagnetic interactions.

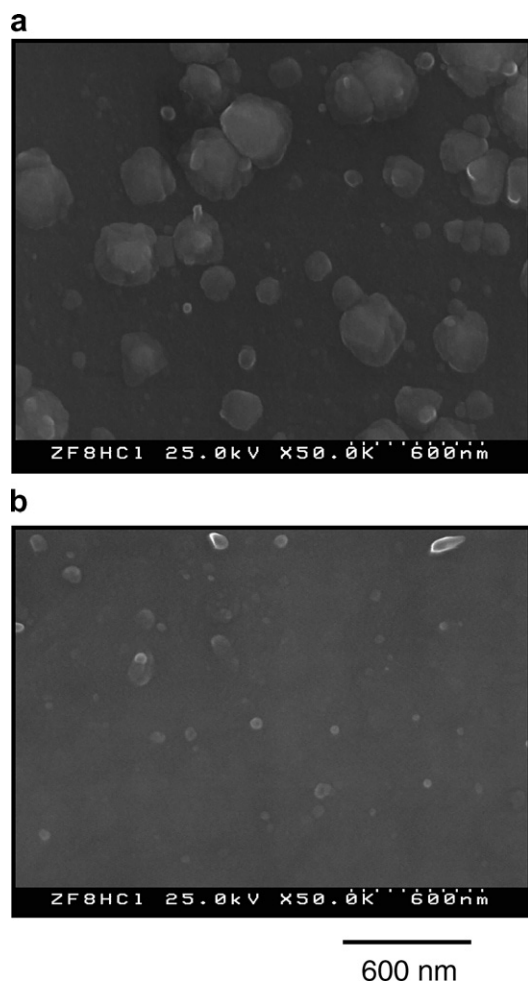


Fig. 13. SEM micrographs of 10ZnFe-800 °C sample. a – free surface; b – fracture surface.

Isothermal magnetization measurements were taken at different temperatures in magnetic fields up to 2 T (Fig. 16). Anhysteretic curves were obtained at 1.7 K and the slight curvature shown agrees well with the presence of the small ferromagnetic interactions revealed by the $\chi^{-1}(T)$ fit. No saturation is found even with an applied magnetic field of 12 T. At 10 K, the perfect linearity of the curve suggests that the system is already paramagnetic at such low temperatures. In agreement with these results the AC-susceptibility measurements taken at 995 Hz in the 1.7–300 K range reveal no transition down to 1.7 K (Fig. 17). Any undetected transition should be very weak since the lock-in sensitivity is able to detect at 43 K the typical anomaly due to contamination of the *MagLab* plastic sample holder by absorbed oxygen. Considering these magnetization data and the relaxation frequencies of most of the Fe^{3+} magnetic moments detected at 4 K by Mössbauer spectroscopy one concludes that T_B of the zinc ferrite particles in the 10ZnFe-800 °C sample is lower than in the 10ZnFe-500 °C samples in agreement with the lower average size of these particles in the former sample. Another possible explanation of the magnetization and 4 K Möss-

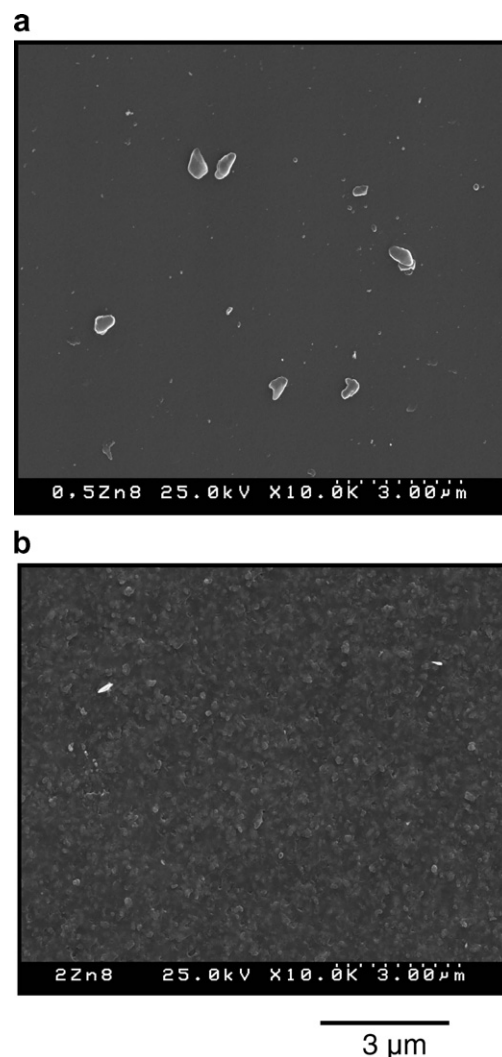


Fig. 14. SEM micrographs of 5ZnFe-800 °C sample. a – free surface; b – fracture surface.

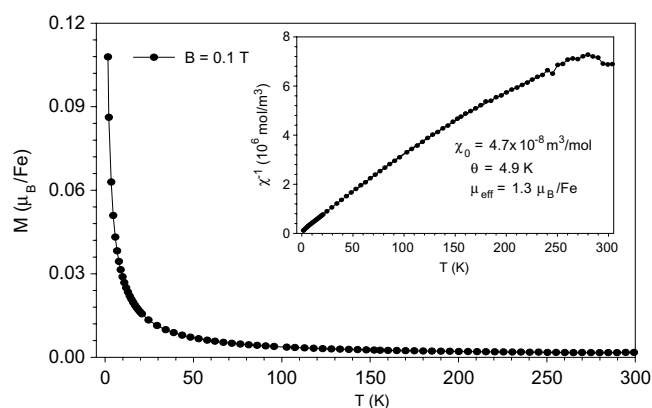


Fig. 15. Temperature dependence of the magnetization, $M(T)$, and the inverse of the magnetic susceptibility curve, $\chi^{-1}(T)$ (inset), of the 10ZnFe-800 °C sample measured at 0.1 T.

bauer data of the 10ZnFe-800 °C is the absence of any kind of magnetic ordering in this sample down to 1.7 K, the low

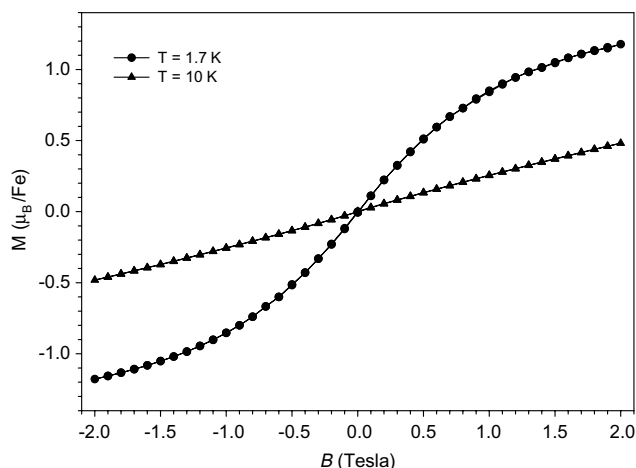


Fig. 16. Magnetic field dependence of the magnetization, $M(B)$, of the 10ZnFe-800 °C sample, taken at 1.7 K and at 10 K.

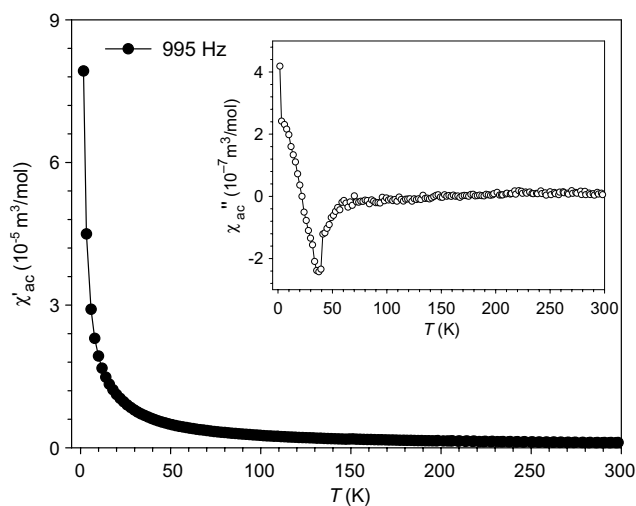


Fig. 17. Temperature dependence of the in-phase $\chi'(T)$ and the out-of-phase, $\chi''(T)$ components of the AC-susceptibility of the 10ZnFe-800 °C sample, taken at 995 Hz with an AC-amplitude of 0.1 mT.

relaxation frequencies of the Fe^{3+} resulting from low spin-spin relaxation at 4 K. In either case a partial dissolution at 800 °C of the particles detected in the 500 °C treated samples seem to be the cause of the observed effects.

4. Conclusions

The magnetization and AC-susceptibility study of the 10ZnFe glass-ceramics prepared by the silica sol-gel method and heat treated at 500 °C, has shown that this glass, transparent and ruby color, contains particles with ferro- or ferrimagnetic interactions and superparamagnetic behavior with a blocking temperature of approximately 14 K. According to XRD and Mössbauer spectroscopy

data the reason for this behavior is the presence of nanoparticles of zinc ferrite with an approximately average size of 24 nm, as determined by the Scherrer method. The zinc ferrite particles in 5ZnFe glasses are much smaller, approximately 6 nm. On the other hand higher concentrations of $\text{ZnO} + \text{Fe}_2\text{O}_3$ in the 20ZnFe samples lead to the formation of a brown and opaque material containing larger hematite and zinc ferrite particles.

Heating at 800 °C seems to cause partial dissolution of the particles and loss of the useful magnetic properties observed in the 10ZnFe glass. As detected by SEM, the average dimension of the particles present in this sample is approximately 6 nm, significantly lower than at 500 °C, and the magnetization study of the glass material reveals a paramagnetic behavior down to 1.7 K.

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